Hard AbsTopK input gate on MNIST: hard backward against DUST steps at five population sizes

2026-10-22

1 Setup

Two-layer MLP on MNIST, $z = W_2\,\mathrm{ReLU}(W_1 x' + b_1) + b_2$ with hidden width 1024, where $x' = \mathrm{AbsTopK}_K(e^g \odot x)$ keeps the K entries of largest magnitude of the rescaled input; $g \in \mathbb{R}^{784}$ is trainable from 0, pixels are in $[0, 1]$. AdamW, learning rate 10^{-3} with 100 warmup steps and a cosine decay to 10^{-4} over 5000 steps of batch 128, no weight decay, no gradient clipping, seed 1337 (the same data order and initialization for every run of a given K).

Hard: the backward passes the hard mask, so g_n moves only through samples whose support contains pixel n. **DUST**: the same gate, with every second optimizer step a DUST step (schedule $[1, 1]$): the gate input $u = e^g \odot x$ of every sample is perturbed on all 784 pixels with $\mathcal{N}(0, \sigma^2)$ noise, $\sigma = 0.1$, in P draws (the population), the tail is rerun, and

$$\hat{g}_t = \frac{1}{P\,\sigma\,B} \sum_{i=1}^{P} \big(\ell_t^{(i)} - \bar{\ell}_t\big)\, a_t^{(i)}, \qquad \hat{G}_g = \sum_t \hat{g}_t \odot e^g \odot x_t$$

is applied to g by the shared optimizer; nothing else moves on a DUST step, so a DUST run has 2500 hard steps for W_1, W_2 against 5000 for the hard run. $K \in \{2, 4, 8, 16, 32\}$, $P \in \{128, 512, 2000, 8000, 32000\}$. Training CE is the clean hard forward on both kinds of step; test CE and accuracy are the deterministic hard forward on the 10 000 test images. The $K \geq 4$ runs were trained on one 4×RTX 3090 box, the $K = 2$ runs on another of the same type with the identical training configuration (those runs also save the extra checkpoints of the SNR study).

2 Results

Figures 1 to 5 and Tables 1 to 5 hold the measurements; all 30 runs finished.

Small K. At $K = 2$, 4 and 8 every DUST run beats the hard backward by a wide margin: test accuracy 60.2–62.4% against 44.0% at $K = 2$ (test CE 1.08–1.15 against 1.56), 75.3–76.3% against 67.1% at $K = 4$ (0.69–0.72 against 0.96), and 86.9–88.0% against 84.6% at $K = 8$ (0.37–0.39 against 0.48). The gap widens as K shrinks. The hard run's training and test CE are flat from about step 1500 (at $K = 2$ its best test CE is at step 600) while the DUST runs keep descending. Under the hard backward g_n moves only through samples that already select pixel n, so the support is fixed early; the DUST steps give every pixel a gradient and the support keeps changing.

Large K. At $K = 16$ and $K = 32$ the test accuracies of all six runs lie within 0.5 points of each other (93.3–93.9% and 95.8–96.3%). The DUST runs reach a lower test CE (0.197–0.213 against 0.218 at $K = 16$; 0.123–0.132 against 0.140 at $K = 32$) with a higher training CE: at $K = 32$ the

hard run's training CE is 0.022 against its test CE of 0.140, while the DUST runs sit at 0.07 and 0.12. Two effects are mixed here: the DUST runs have 2500 hard steps for W_1, W_2 against 5000, and their support keeps moving; this sweep does not separate them.

Population. The gain is already present at $P = 128$ (two tail forwards of batch 128 per DUST step) and grows little with P: from 128 to 32 000 the test accuracy moves by about two points at $K = 2$, one point at $K = 4$ and $K = 8$ and within noise at $K = 16$ and $K = 32$, and the training CE of the DUST runs drops by 0.01–0.1 (the most at $K = 2$). At this problem size the estimator's variance is not what limits the method; the cost of a DUST step is linear in P (0.8 s at $P = 32\,000$ against 12 ms at $P = 512$ on an RTX 3090).

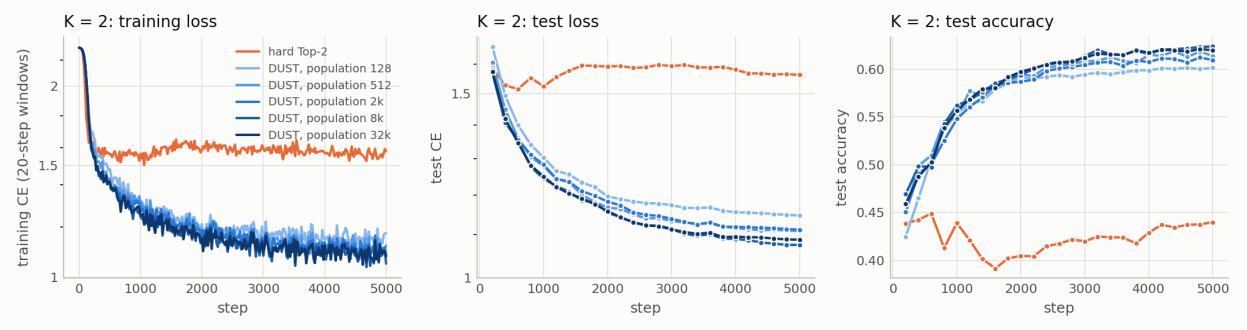


Figure 1: $K = 2$: training CE (left, log scale, 20-step window means), test CE (middle, log scale, every 200 steps) and test accuracy (right) for the hard backward and DUST at five populations.

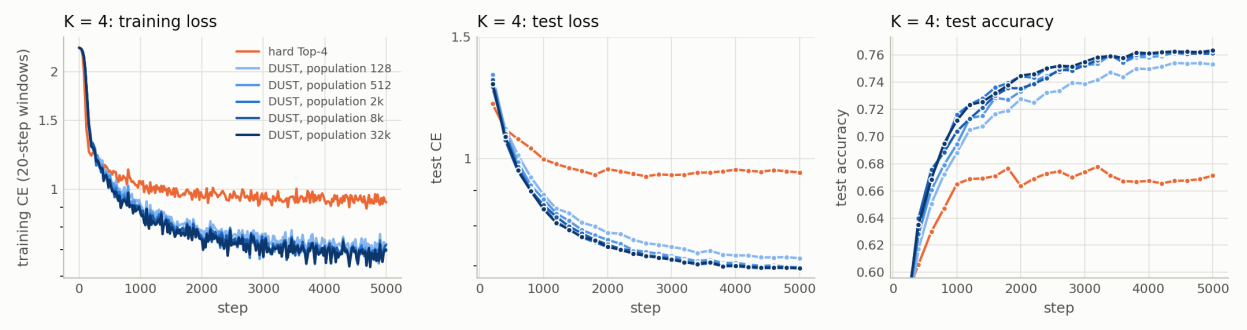


Figure 2: $K = 4$: training CE (left, log scale, 20-step window means), test CE (middle, log scale, every 200 steps) and test accuracy (right) for the hard backward and DUST at five populations.

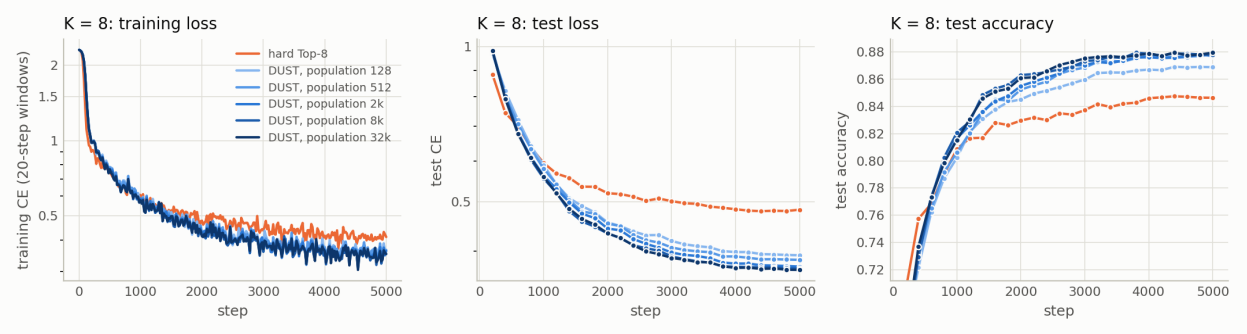


Figure 3: $K = 8$: training CE (left, log scale, 20-step window means), test CE (middle, log scale, every 200 steps) and test accuracy (right) for the hard backward and DUST at five populations.

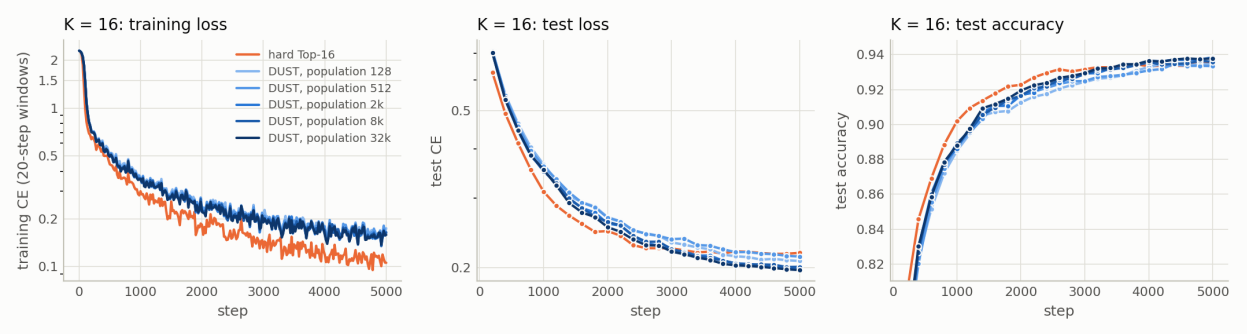


Figure 4: $K = 16$: training CE (left, log scale, 20-step window means), test CE (middle, log scale, every 200 steps) and test accuracy (right) for the hard backward and DUST at five populations.

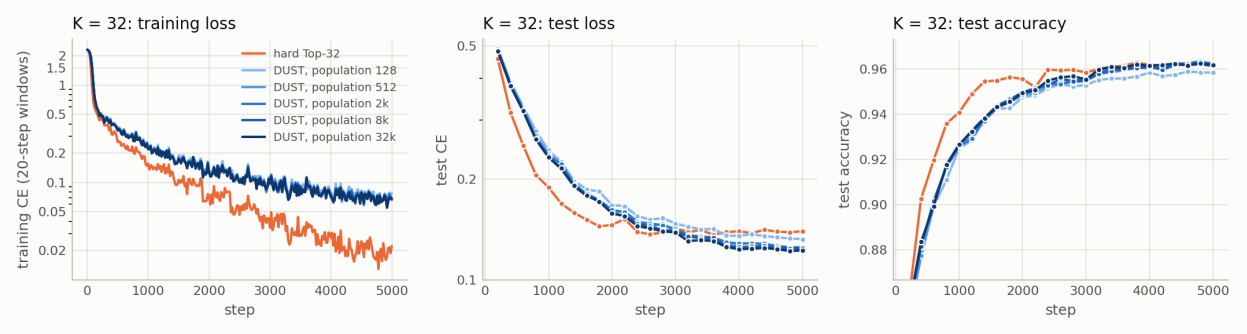


Figure 5: $K = 32$: training CE (left, log scale, 20-step window means), test CE (middle, log scale, every 200 steps) and test accuracy (right) for the hard backward and DUST at five populations.

Table 1: $K = 2$: values at the final step (5000). Training CE is the mean of the last 20-step window of the clean hard forward; test CE and accuracy are the deterministic hard forward on the 10 000 test images.

method	population	train CE	test CE	test acc (%)
hard Top-K	–	1.5802	1.5629	44.02
DUST	128	1.1730	1.1477	60.15
DUST	512	1.1223	1.1090	61.41
DUST	2,000	1.0802	1.1120	60.97
DUST	8,000	1.0509	1.0754	62.40
DUST	32,000	1.0792	1.0878	61.98

Table 2: $K = 4$: values at the final step (5000). Training CE is the mean of the last 20-step window of the clean hard forward; test CE and accuracy are the deterministic hard forward on the 10 000 test images.

method	population	train CE	test CE	test acc (%)
hard Top-K	–	0.9261	0.9556	67.12
DUST	128	0.7252	0.7180	75.31
DUST	512	0.7025	0.6972	76.06
DUST	2,000	0.6957	0.6963	76.10
DUST	8,000	0.7207	0.6943	76.14
DUST	32,000	0.6998	0.6943	76.34

Table 3: $K = 8$: values at the final step (5000). Training CE is the mean of the last 20-step window of the clean hard forward; test CE and accuracy are the deterministic hard forward on the 10 000 test images.

method	population	train CE	test CE	test acc (%)
hard Top-K	–	0.4113	0.4822	84.64
DUST	128	0.3657	0.3932	86.89
DUST	512	0.3630	0.3852	87.84
DUST	2,000	0.3626	0.3738	87.70
DUST	8,000	0.3518	0.3704	87.81
DUST	32,000	0.3515	0.3683	87.96

Table 4: $K = 16$: values at the final step (5000). Training CE is the mean of the last 20-step window of the clean hard forward; test CE and accuracy are the deterministic hard forward on the 10 000 test images.

method	population	train CE	test CE	test acc (%)
hard Top-K	–	0.1057	0.2181	93.50
DUST	128	0.1745	0.2080	93.43
DUST	512	0.1646	0.2135	93.34
DUST	2,000	0.1642	0.1985	93.85
DUST	8,000	0.1643	0.2010	93.58
DUST	32,000	0.1586	0.1972	93.78

Table 5: $K = 32$: values at the final step (5000). Training CE is the mean of the last 20-step window of the clean hard forward; test CE and accuracy are the deterministic hard forward on the 10 000 test images.

method	population	train CE	test CE	test acc (%)
hard Top-K	–	0.0223	0.1399	96.14
DUST	128	0.0775	0.1324	95.84
DUST	512	0.0687	0.1226	96.26
DUST	2,000	0.0687	0.1251	96.22
DUST	8,000	0.0683	0.1245	96.14
DUST	32,000	0.0666	0.1227	96.18